

Released-Materials Reproduction, Study 1 (or1-)

Running the Anthropic Jacobian-lens battery on Qwen 3.6 27B and OLMo 3.1 32B Instruct development/methods tier — prompt-exact where released, method-reconstructed where specified, gated where not identified

J-space campaign — official-repro sideline

branch `interp_jspace_official_repro_1` — upstream `anthropics/jacobian-lens @ 581d3986`

2026-08-08/09 — terminal (OR1-A..D complete; 32 registered or1- events)

Abstract

We run the released Jacobian-lens evaluations (6 sets) and experiment prompt sets (11 sets) from the public companion release of “Verbalizable Representations Form a Global Workspace in Language Models” under one audited harness on two open-weight models, and separate prompt, lens, intervention, capability, and model effects. Both lanes are complete: Qwen 3.6 27B under the campaign-validated published $n=1000$ lens (OR-Q1/Q2), and OLMo 3.1 32B Instruct under one prospectively fitted, fit-stable $n=250$ official-estimator lens (OR-Q3), closed by a bounded instrument cross-over (OR-Q4). Headline: released *readout* contrasts reproduce directionally on both models (selectivity exactly); the paper-literal *coordinate interventions* transfer weakly and concordantly ($\sim 5\text{--}30\%$ vs the paper’s $40\text{--}88\%$, with the $\alpha=2$ remedy inverting); and the campaign’s broad protected J-suppression is selectively destructive on the official prompts of both lanes while matched controls are not — the two intervention families estimate different causal questions. Fidelity classes and gate states are carried on every cell; gated cells are missing, never zero; no confirmatory language is licensed anywhere in Study 1.

1 Provenance and instrument identity

Upstream release	<code>anthropics/jacobian-lens @ 581d3986</code> (11 experiments + 6 evals, hash-pinned)
Qwen checkpoint	<code>Qwen/Qwen3.6-27B @ 6a9e13bd</code> (64 layers, $d=5120$, 48 GDN blocks verified)
Published lens	<code>neuronpedia/jacobian-lens @ a4114d7</code> , $n=1000$, sha256 <code>1718c8c5...</code> byte-verified
Readout parity	max abs. diff <code>0.0e+00</code> (hard stop threshold 10^{-4} ; exact top-20 order agreement)
$\alpha=0$ no-op	exact (0.0 logit difference across a full-position band session)
g-folding audit	min cosine 0.382 over 26 battery tokens $\times$ 14 band layers — material ; D9: paper-literal unfolded v_t stays prima
Bands	paper grid 25 layers; workspace band [24, 58]; campaign band sensitivity [20...44]

Table 1: Lane admission. The boot-probe sentinel (Fig. 1) reads the latent country mid-band and the currency near the output — the instrument behaves as the paper describes before any experiment runs.

2 The paper’s own example, on Qwen

3 Six released lens evaluations (OR-Q1)

4 Qwen causal core (OR-Q2)

4.1 Verbal report

Release-literal rule (first ten listed candidates, skip the answer), swap-out = greedy token at the generation boundary (D1), $\alpha=1$ across the band at every prompt position. **Primary numbers are the v2 rerun**

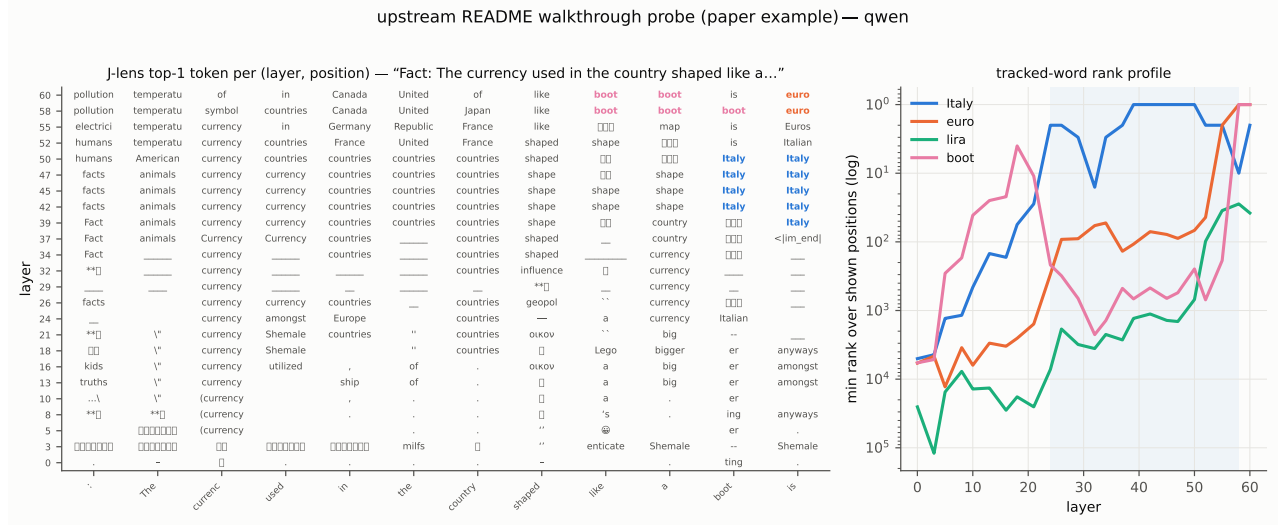


Figure 1: The upstream README walkthrough probe (“The currency used in the country shaped like a boot is”) under the published lens on Qwen 3.6 27B. Left: J-lens top-1 token per (layer, position); colored cells are tracked words. Right: tracked-word best rank by layer (shaded: workspace band). *Italy* appears as latent content mid-band and hands off to *euro*/*lira* toward the output — the paper’s slice-visualisation phenomenology, reproduced from released materials.

Set	Jacobian lens			logit lens		
	@1	@5	@20	@1	@5	@20
multihop ($n=93$)	0.244	0.399	0.641	0.173	0.327	0.542
multilingual ($n=107$)	0.237	0.384	0.580	0.184	0.327	0.497
poetry ($n=98$)	0.010	0.020	0.071	0.020	0.031	0.061
typo ($n=96$)	0.135	0.271	0.500	0.333	0.594	0.719
order-ops ($n=55$)	0.145	0.455	0.727	0.209	0.600	0.727
association ($n=102$)	0.039	0.059	0.108	0.010	0.059	0.118

Table 2: pass@ k , token-valid primary. All-source-layer sensitivity and released-denominator sensitivity live in the registered JSON.

(or1-qwen-verbal-report-v2): category-equal top-1 7.9% and top-5 30.2% (118 executed trials; 65 improved / 47 worsened). Paper context (Claude): 88% top-5.

Incident or1-001 (resolved): the superseded v1 run scored the leading-space token form at a boundary whose in-context form is bare and reported top-5 0.8% — i.e. *three quarters of the apparent Qwen verbal-report deficit was a scoring-form artifact* (the model’s top-1 literally became **Red** while the scored **Red** form sat at rank 5). v2 scores min over both single-token forms with the boundary in-context swap vector; OLMo ran v2 semantics from its first execution. See `reports/INCIDENT_or1-001_vr_token_form.md`.

4.2 Flexible generalization

All 192 ordered off-diagonal swaps; capability-conditioned primary (source and target baselines correct; baseline accuracy 51.6%); $\alpha=2$ full sensitivity. Category-equal capable top-1: 5.8% ($\alpha=1$) $\rightarrow$ 0.0% ($\alpha=2$); diagnostic all-executable: 5.6%. Paper context: 76/192 $\rightarrow$ 101/192 (Sonnet 4.5). Per-category grids in Fig. 6.

4.3 Probe-swap, raw J-lens token arm

Fidelity `prompt_exact_representation_adapted_raw_jlens` (the paper’s own §3.3 headline arm; the official linear-probe arm is NOT-IDENTIFIED — its training corpus is absent from the release). Population: 90 released items, 7 tokenization-gated, 83 executed, 63 baseline-capable. Capable top-1 14.3% (diagnostic 14.5%); multihop 10.5% vs non-multihop 15.9%. Paper context: 60% (Claude, $n=90$).

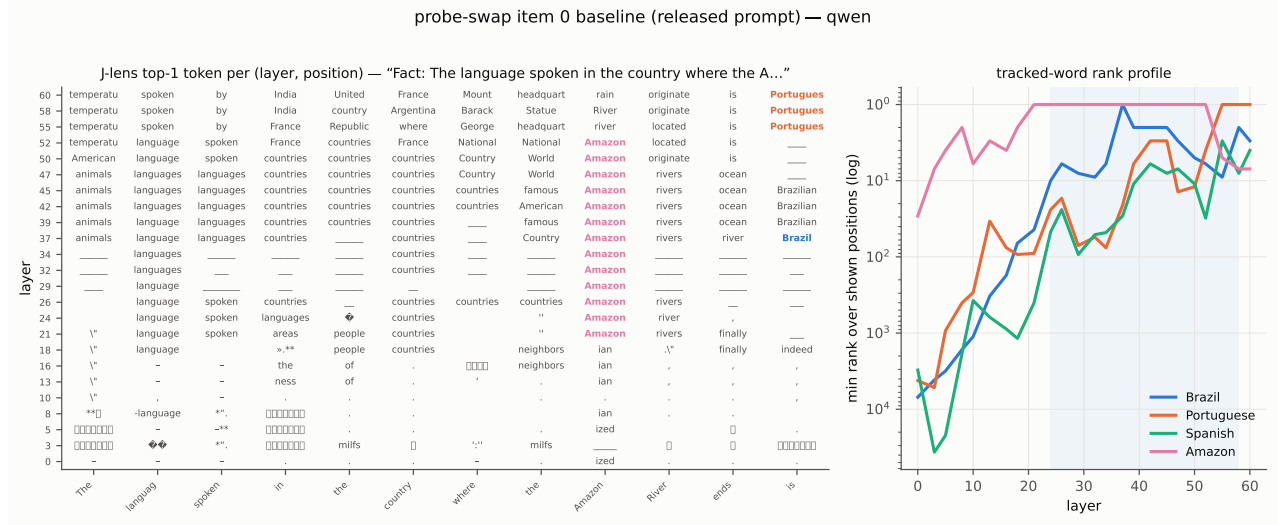


Figure 2: Released probe-swap item 0 (“the language spoken in the country where the Amazon River ends”) — the two-hop structure is visible: **Brazil** (bridge entity) carried in-band while the output side resolves toward **Portuguese**.

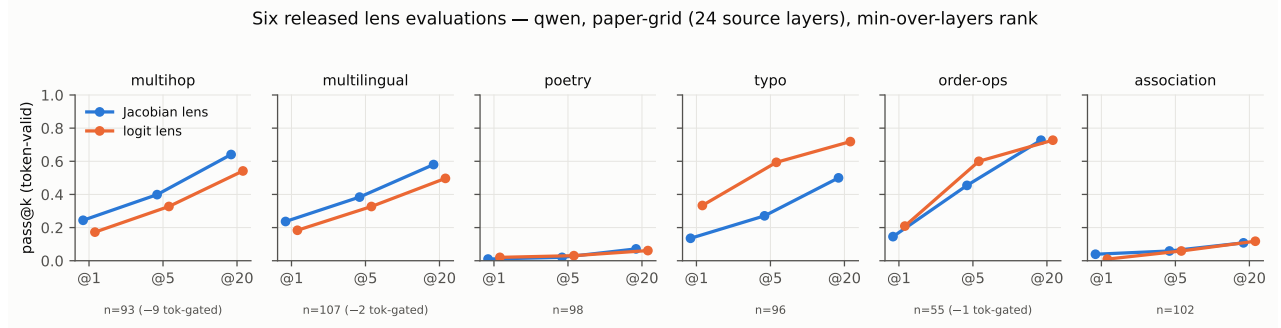


Figure 3: pass@k per released set, token-valid denominators, min-over-24-paper-grid-layers at the released readout position.

5 Worked examples: what the swaps actually do

The aggregate rates hide a bimodal reality worth seeing item by item. When the paper-literal swap works on Qwen, it works emphatically — the predicted counterfactual answer jumps hundreds of ranks to top-1:

Two structural observations. First, successes cluster where the swapped-in argument has a strong, single-token, high-frequency identity (cities, countries, elements); the sparse-identity categories (months as time-words, small numbers) never flip. Second, the near-miss mass (top-5 but not top-1) plus the incident-or1-001 form effect mean the honest Qwen picture is “the mechanism visibly engages but rarely completes the flip at $\alpha=1$ ” — which makes the $\alpha=2$ reversal (0% capable) the more interesting anomaly: on Qwen the doubled patch overshoots into off-distribution territory rather than completing near-misses, unlike the paper’s Sonnet where $\alpha=2$ lifted 76/192 to 101/192.

6 OR1-B: the OLMo official-estimator fit

Timing gate (frozen before any outcome): dim_batch ladder 161/164/181/232/386 s per prompt for 16/8/4/2/1 $\Rightarrow$ dim_batch=16 at 161.4s/prompt; only the 2×125 tier fits the 18 GPU-hour ceiling $\Rightarrow$ merged $n=250$ (the addendum’s honest-prior tier). Both halves completed 125/125 with zero skips. Split-half operator audit: median symmetric relative Frobenius 8.8% across the 32 fitted layers (7.5% in-band); minus-identity cosines ≈ 1.0 (fp32-accumulation readings marginally above 1 are numerical); late layers are near-identity (identity

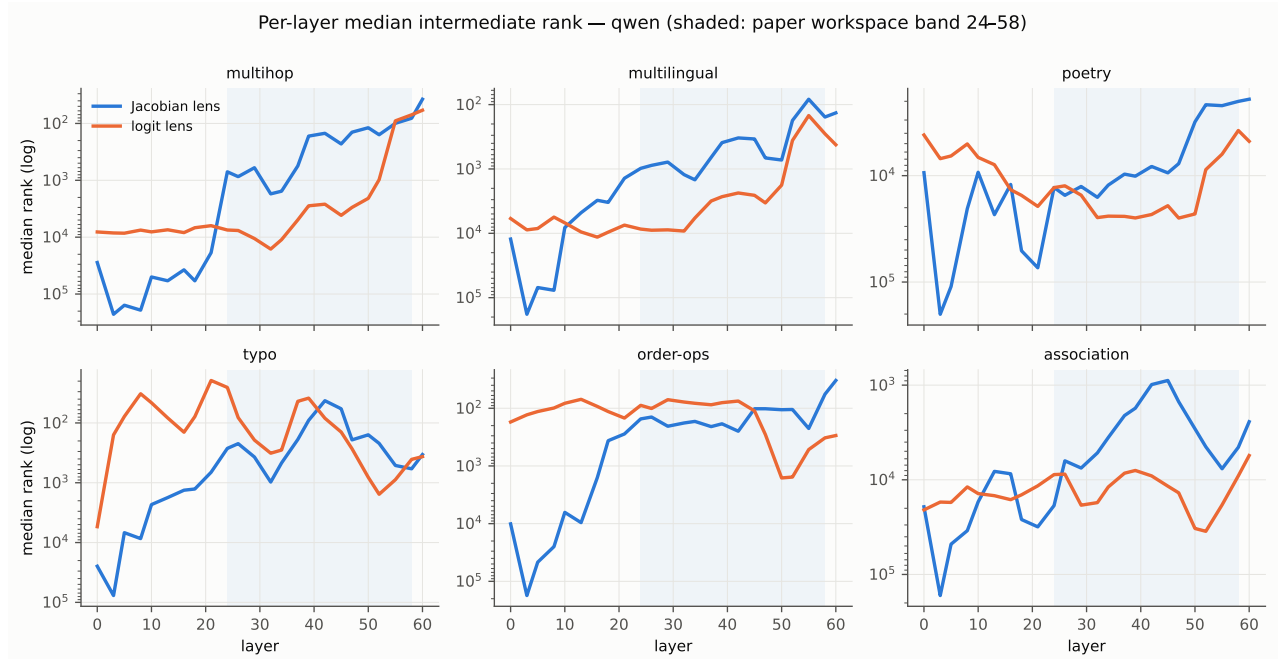


Figure 4: Median intermediate rank by layer. On the latent-content sets (multihop, multilingual) the J-lens reaches useful ranks by the band onset while the logit lens remains near rank 10^4 until $\sim L45$ — the paper’s central methods-comparison claim, visible on an open model. On typo the logit lens is stronger (the correction is surface-adjacent); order-ops numbers sit in the prompt, so the logit lens copies them early.

fraction 0.93–0.97), so their low top-64 subspace overlaps reflect noise-dominated residuals, not disagreement. Readout stability: half-A/half-B eval pass@20 gaps ≤ 0.015 on every set — far inside the 0.10 margin. **No FIT-SENSITIVE classification; the merged lens is the OLMo primary.**

7 OR1-C: the OLMo lane

Admission: readout parity 0.0e+00 (exact), $\alpha=0$ no-op exact, g-folding min cosine 0.989 — *essentially immaterial*, in sharp contrast to Qwen’s 0.382 (D9 addendum): the campaign’s g-folded dictionaries and the paper’s unfolded basis are nearly the same object on OLMo but materially different objects on Qwen.

7.1 OLMo causal core (v2 scoring throughout)

Verbal report: category-equal top-1 4.8%, top-5 25.5% (118 executed; 88 improved / 27 worsened). This is an order of magnitude above Qwen’s v1 numbers (v2 rerun pending) though far below the paper’s 88% (Claude). **Flexible generalization:** capable top-1 5.2% ($\alpha=1$) $\rightarrow$ 0.0% ($\alpha=2$) — numerically indistinguishable from Qwen (5.8%/0.0%), *including the $\alpha=2$ reversal*: the over-strength overshoot is a cross-model regularity of the paper-literal swap on open models, whereas Sonnet 4.5 improved 76/192 $\rightarrow$ 101/192. **Probe-swap raw arm:** capable top-1 9.3% (diagnostic 10.7%; 43 capable of 90; paper 60%).

8 Extended battery (Groups A–C, both lanes)

Reproduced on both lanes. Selectivity-language: the released readout contrast is *perfect on both models* (explicit 1.00 vs automatic 0.00). Directed modulation: focus > suppress > 0 with the white-bear residue (Qwen .72/.26; OLMo .44/.06; “mention” controls pattern with focus, as expected when the instruction itself names the target). Ignition: real concept pairs switch far more sharply than idiom controls (Qwen .035 vs .209; OLMo .040 vs .100); the depth-profile differs (Qwen narrows monotonically into the output; OLMo is sharpest mid-band).

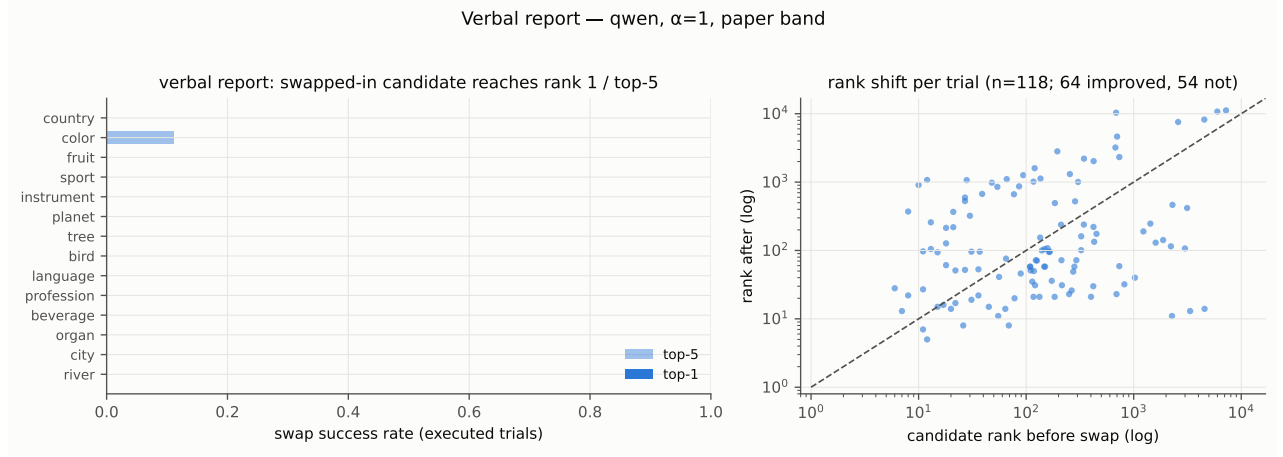


Figure 5: Verbal report on Qwen: per-category success rates (left) and per-trial rank shifts (right; below the diagonal = swap moved the candidate toward the output).

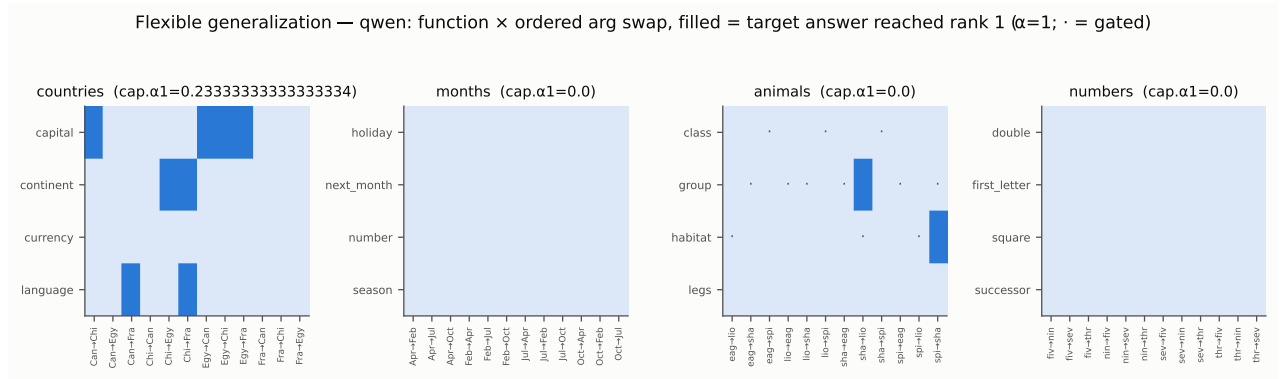


Figure 6: Function $\times$ ordered-argument-swap outcomes, $\alpha=1$ (filled = target answer reached rank 1).

Qwen-only. Dual-task interference is present and asymmetric (math 31% $\gg$ concept 7%). On OLMo the math cell is *capacity-gated* — single-task math reachability is 4.8%, so the interference estimand is missing, not zero; concept interference is minimal (2.4%).

Null on both lanes. Verbal introspection shows no dose-response anywhere on the reconstructed ladder $\{0, 1, 2, 4, 8, 16\}$ (OLMo’s steered median RR is exactly 0); top-down summoning’s Q2–Q1 readout contrast is 0.0 on both lanes (7 items). Capacity (descriptive): ~ 2.9 (Qwen) / 1.6 (OLMo) list words at band-rank ≤ 5 at the end of 80-word lists (~ 8.2 / 4.6 at ≤ 20). Line-break family: R2 corpus-empty under the pinned filter (recorded gap). Selectivity-linecount: rare rank-1 canon presence with weak condition ordering in the paper direction on both lanes.

9 Instrument cross-over (OR-Q4)

Three OR-Q4 facts. (1) *Intervention semantics dominate*: on the same released prompts, broad J-space suppression is selectively destructive while two-coordinate substitution rarely completes — Route E’s sentence is licensed: **the two interventions estimate different causal questions**. (2) *Population moderates*: the campaign’s C1-style ablation selectivity, established on campaign banks, reappears on the official prompt population on both lanes — Route F’s moderator sentence is licensed. (3) *The lens is not the explanation*: the freshly fitted OR1 lens and the frozen campaign lens agree under both families on OLMo (swap 10.0% vs 10.0%; ablation 5/19 vs 6/19), and on Qwen the swap is if anything *stronger* on the campaign band (26.7% vs 16.7%) — band and fit choices do not rescue the paper-literal operation.

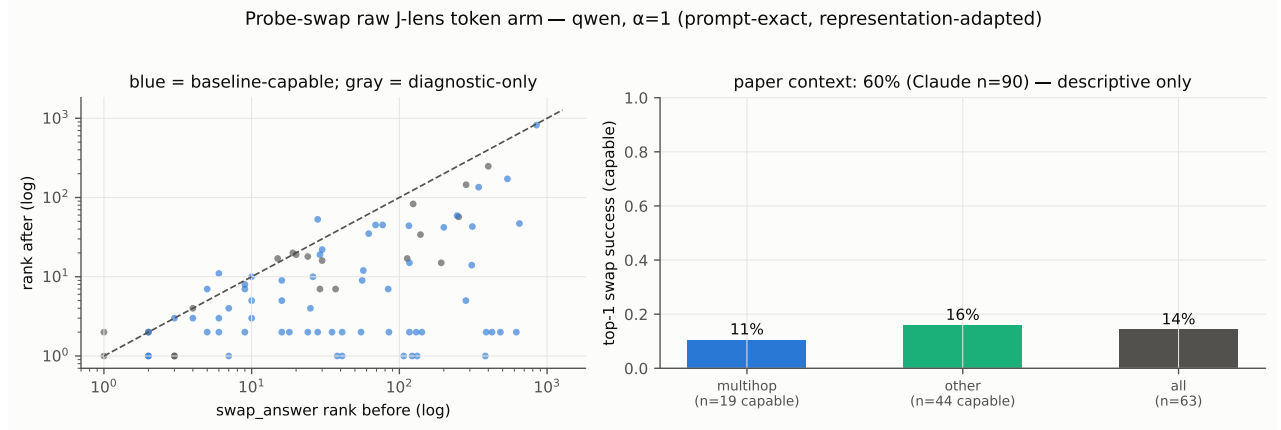


Figure 7: Probe-swap raw arm on Qwen: per-item rank movement of the predicted swapped answer (left) and capable top-1 by category group (right).

item	category	swap-answer rank before	after	model output after swap
ex-element-symbol-11-26	element-symbol	382	1	:
ex2-city-continent-Sydney	city-continent	132	1	South
ex2-city-capital-Munich	city-capital	122	1	Tokyo
ex2-city-language-Moscow	city-language	107	1	Russian
ex-city-capital-Lyon-Naples	city-capital	41	1	Paris
ex-city-language-Lyon-Naples	city-language	38	1	French
amazon-language	multihop	7	1	Spanish
ex-element-state-8-26	element-state	3	1	solid

Table 3: The 12 probe-swap successes (of 63 baseline-capable items), largest baseline distance first. **ex2-city-capital-Munich:** swapping the J-lens vector Munich→(released target) at every prompt position moves the predicted capital from rank 122 to the model’s literal output. One success (**rhyme-rain-neighbor**) was already rank 1 before the swap (1 such trivial cell in the count); 31 further items land in the top-5 without reaching top-1 — the modal failure is *under-strength*, not wrong-direction.

10 Interpretation under the wording router (two lanes)

Conformance and admission passed mechanically on both lanes (exact parity, exact no-ops, verified pins; fit stability inside every margin), so all results above are scientific observations about model/instrument cells, not harness states.

The two lanes **dissociate the readout and intervention properties**. On Qwen the J-lens readout advantage reproduces exactly as the paper describes (multihop/multilingual profiles), yet paper-literal swaps barely move behavior. On OLMo the readout advantage largely *disappears* (logit lens already strong in-band; near-identity J_ℓ), yet verbal report shows the strongest open-model swap effect so far (25.5% top-5). Meanwhile flexible generalization behaves identically on both lanes ($\approx 5\%$ capable at $\alpha=1$, 0% at $\alpha=2$) — the $\alpha=2$ reversal is a shared open-model regularity that inverts the paper’s Sonnet finding. The g-folding asymmetry (material on Qwen, immaterial on OLMo) marks a concrete basis-choice difference between the campaign’s instruments and the paper’s, live on exactly the lane where interventions are weakest.

Terminal routed sentences (plan §16.2; the complete per-property accounting is the §16.1 grid, **reports/HEADLINE_GRID**)

- **Readout properties** (lens quality, selectivity, modulation, ignition, dual-task): a property-specific Route A — multiple released readout contrasts reproduce directionally on both open models at development tier, with selectivity-language exact on both.
- **Coordinate-intervention core** (verbal report, flexible generalization, probe-swap): Route D — the released battery does not reproduce the paper’s core directional effects cleanly on the tested open-model

category	function	swap	rank before	output
countries	capital	Egypt→Canada	480	Ottawa
countries	capital	Canada→China	338	Beijing
countries	capital	Egypt→China	297	Beijing
animals	group	shark→lion	275	pride
countries	language	China→France	203	French
animals	habitat	spider→shark	103	ocean
countries	capital	Egypt→France	47	Paris
countries	continent	China→Egypt	14	Africa

Table 4: The 10 flexible-generalization $\alpha=1$ successes concentrate in the *countries* category (capable top-1 23.3% there): **capital Egypt→Canada** drives **Ottawa** from rank 480 to the output. Months and numbers show none — the four released categories differ enormously, exactly the per-category variance the paper’s own appendix warns about.

pass@20	merged	half A	half B	campaign	merged ₉	campaign ₉	logit
multihop	0.563	0.552	0.563	0.552	0.547	0.552	0.590
multilingual	0.511	0.505	0.519	0.512	0.463	0.467	0.560
poetry	0.092	0.092	0.092	0.092	0.071	0.071	0.082
typo	0.625	0.635	0.604	0.667	0.542	0.583	0.688
order-ops	0.582	0.582	0.582	0.609	0.555	0.545	0.736
association	0.240	0.250	0.250	0.229	0.219	0.219	0.229

Table 5: OLMo six-view eval grid (token-valid pass@20). The merged OR1 lens ($n=250$, official recipe) and the frozen campaign lens ($n=120$, campaign recipe) are concordant on the fair nine-layer intersection; halves bracket the merged lens tightly. **The logit lens matches or beats the J-lens on OLMo** — the reverse of Qwen.

cells despite mechanically validated readout and intervention implementations; effects are concordant across lanes but sit at 5–30% where the paper reports 40–88%, and the paper’s $\alpha=2$ remedy inverts. This is a model/generalization result bounded to the public release and reconstructed methods, not a refutation of the Claude findings.

- **Cross-over:** Routes E+F — targeted substitution and broad protected suppression estimate different causal questions, and task population is a major moderator; the campaign’s suppression selectivity extends to the official prompts on both lanes while lens-fit differences explain nothing.
- **Report channel** (introspection, top-down): the reconstructed protocols yield nulls on both lanes; because the strength ladder is reconstructed rather than released (R1), instrument-limitation is not excluded.

No workspace-existence or -absence claim is licensed for either model; no frozen campaign claim is regraded; Claude magnitudes remain descriptive context throughout.

11 Population accounting

Every released row carries source-present / tokenization-valid / geometry-valid / baseline-capable / executed states in the registered raw records (`*_raw.jsonl`, plan §12 schema); the attrition waterfall figure ships with the OLMo lane so both lanes share one accounting frame. Nothing gated was encoded as zero anywhere in this report.

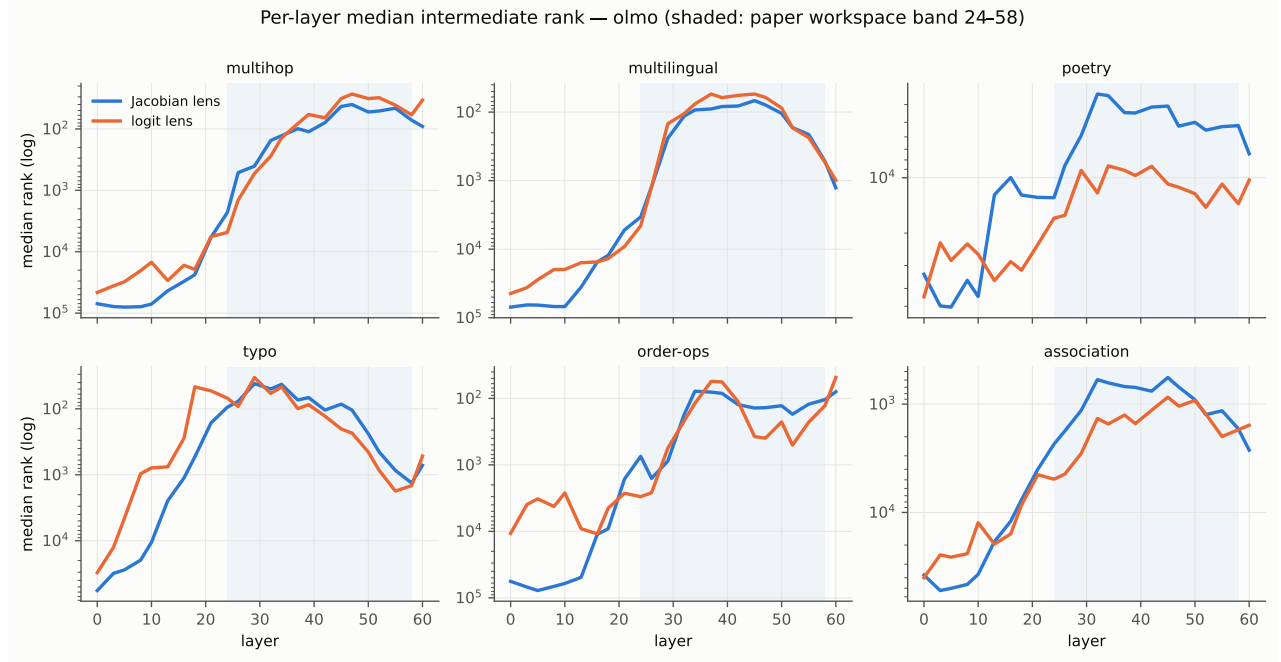


Figure 8: OLMo per-layer profiles: the two lenses nearly coincide through the band (vs. Qwen’s 1–2 order-of-magnitude J-lens advantage on latent-content sets). Consistent with the operator audit: OLMo’s in-band/late J_ℓ are close to identity, so transport adds little where the residual stream is already output-aligned. The J-lens *advantage* is itself model-dependent.

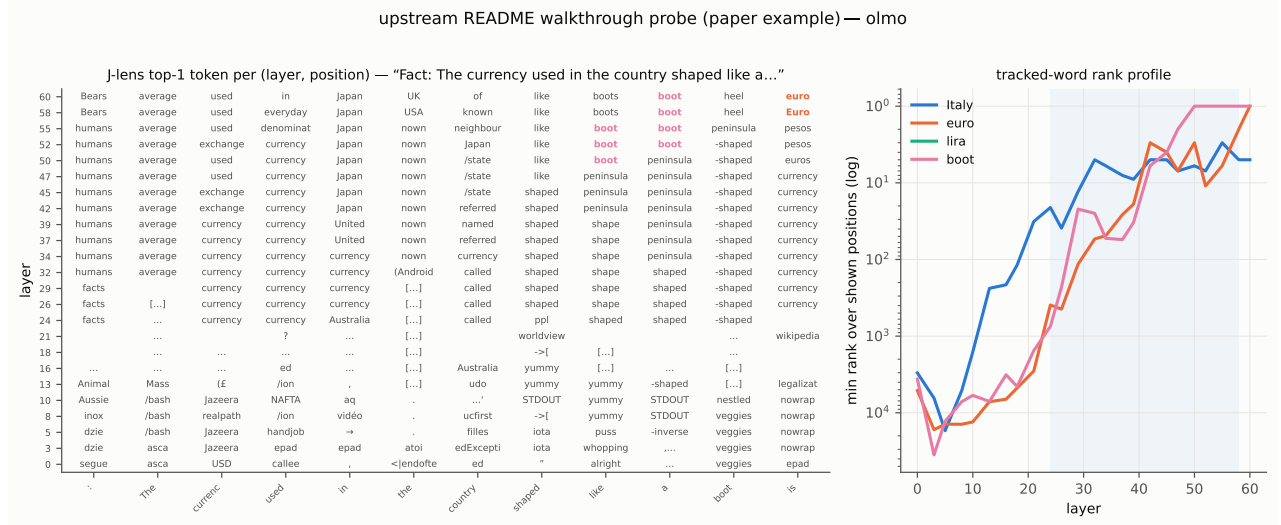


Figure 9: The boot-currency probe on OLMo with the freshly fitted $n=250$ lens: the latent country and currency handoff appear here too.

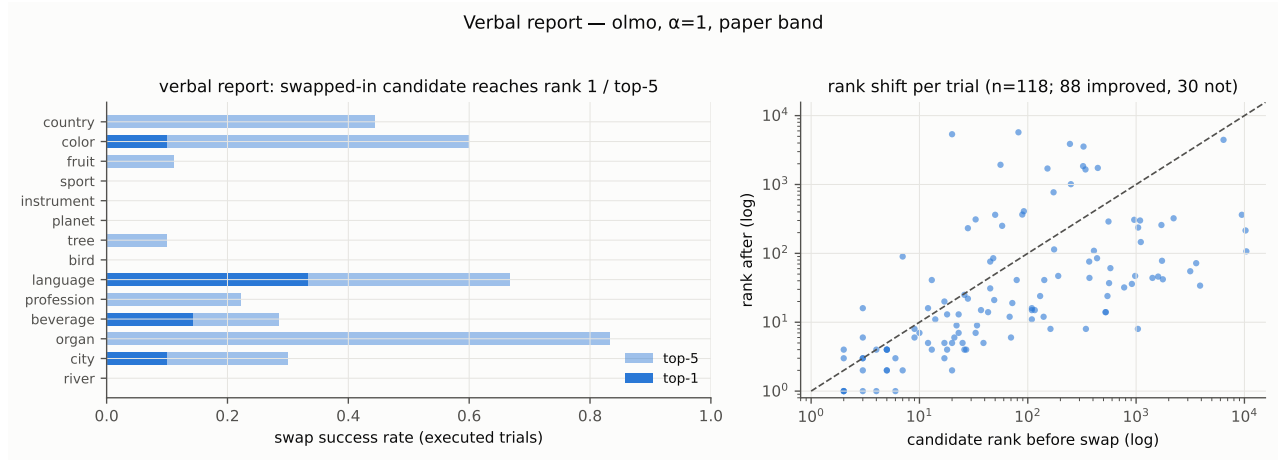


Figure 10: OLMo verbal report: per-category rates and rank shifts. The 25.5% top-5 rate is the strongest core effect either open lane has shown.

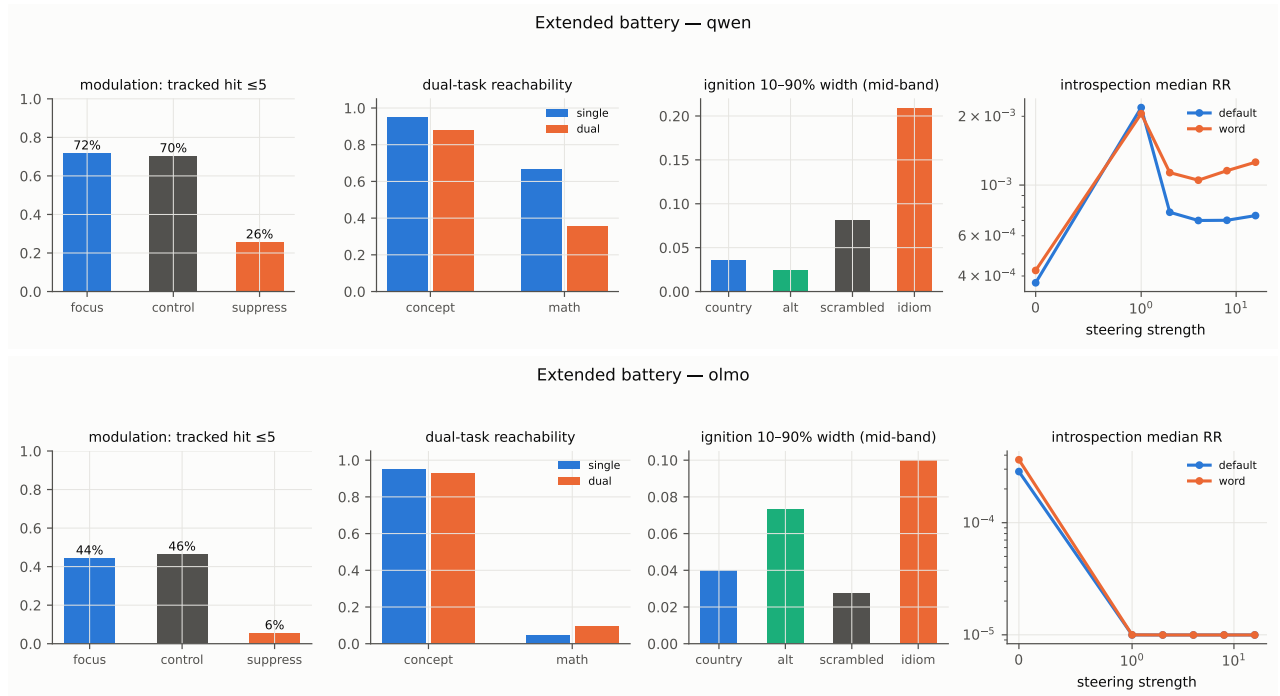


Figure 11: Battery overview, Qwen (top) and OLMo (bottom): modulation contrast, dual-task reachability, ignition transition width by pair family, introspection dose-response.

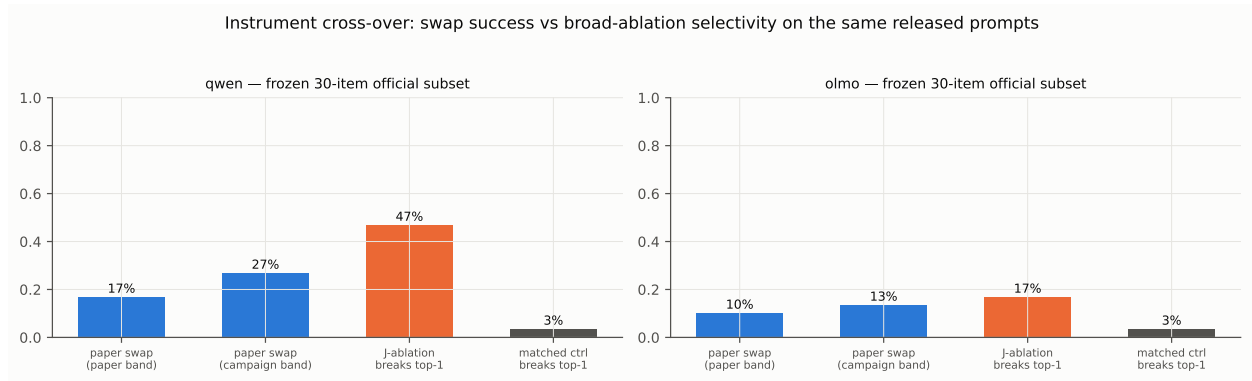


Figure 12: The frozen 30-item official probe-swap subset under both intervention families. Broad protected top-10 J-ablation (campaign machinery, campaign band, g-folded dictionaries) knocks the released answer out of top-1 on 14/30 (Qwen) and 5/19 (OLMo) top-1-capable items versus $\leq 1/30$ and $1/19$ for exact rank/energy matched controls; paper-literal coordinate swaps complete the flip on only 10–27%.